



Figure 1: The Kinect sensor

was founded by PrimeSense, Willow Garage, SideKick and ASUS. OpenKinect publishes its code under Apache 2.0 or GPL 2 licences, while OpenNI publishes its work under different licences. There are mainly two open platforms or libraries, namely *libfreenect* and OpenNI. These have been developed for almost the same purpose, and both support various languages like Python, C++, C#, JavaScript, Java JNI, Java JNA and ActionScript.

Some amazing things/projects that can be done with Kinect:

- ROS (Robot Operating System) Kinect is an open source project focused on the integration of the Kinect sensor with ROS. ROS uses both drivers—OpenNI and *libfreenect*.
- Kinect-controlled computers: Based on user gestures and/or speech recognition, Kinect can control the computer—

although no one has done it for Linux yet.

- Scanning of 3D objects: Kinect enables robots to map 3D objects, resulting in detailed and precise models of people.
- Medical applications: Gesture-based control of surgical tools.
- Education: Writing and calculator applications have already been developed using it.

And there are many more....

In subsequent articles, I plan to cover the installation for *libfreenect* and some interesting Kinect applications (of course, with source code). **END** 

Further reading

- [1] <http://openkinect.org/>
- [2] <http://openni.org/>
- [3] <http://www.ros.org/>

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